Scientific Computing Exercise Set 2

Federico Rossi (16363507) & David Klein Leunk(12992976)

Repository: *https://github.com/DKLEINLEUNK/scientific-computing-2/*

EXERCISE 1

I. Introduction

Diffusion-limited aggregation (DLA) describes the patterns that emerge when particles move randomly and stick upon contact. Starting from a single seed, the cluster grows as diffusing particles reach its boundary. Instead of explicitly tracking random walkers, the diffusion process can be represented by a concentration field satisfying Laplace's equation, with growth probabilities assigned from the local concentration.

In this model a morphology parameter controls how strongly growth favors regions of high concentration. In this exercise we study its effect numerically and explore strategies to improve the computational performance of the simulations.

II. Theory

DLA can be formulated as a boundary-growth process governed by steady-state diffusion. Diffusing particles move toward an existing cluster, while the concentration field is assumed to relax much faster than the cluster grows. The transport field is therefore described by the time-independent diffusion equation (Laplace's equation),

$$\nabla^2 c = 0,$$

where c denotes the concentration of diffusing particles. At each iteration the equation is solved outside the cluster, allowing the concentration near its boundary to be determined.

Growth can occur only at candidate sites adjacent to the cluster. Each candidate (i, j) is assigned a growth probability based on the local concentration $c_{i,j}$,

$$p_g(i, j) = \frac{c_{i,j}^{\eta}}{\sum_{\text{growth candidates}} c_{i,j}^{\eta}}.$$

This normalization ensures that the probabilities over all candidate sites sum to one. A single candidate is then selected according to p_g and added to the cluster. After each growth event the geometry changes, the diffusion field is recomputed by solving Laplace's equation again, and the process is repeated.

The parameter η controls how strongly growth favors regions of high concentration and therefore determines the cluster morphology. For $\eta = 1$ the standard DLA cluster is obtained. For $\eta < 1$ the differences between candidate sites are reduced, leading to more compact structures; in the limiting case $\eta = 0$ all candidates are equally likely and the growth approaches Eden-cluster behavior. For $\eta > 1$, growth mainly occurs at exposed tips where the concentration is highest, producing more open clusters.

III. Methods

The DLA cluster is generated through an iterative procedure in which the steady diffusion equation is solved and the cluster grows by adding one lattice site at a time. The simulations are performed on a square lattice of size $N \times N$ (typically $N = 100$). The cluster is represented by a Boolean grid, initially containing a single seed located near the lower boundary at $(1, N/2)$: slightly offset from the lower boundary to reduce boundary artifacts.

A. Diffusion equation

At each growth step the steady diffusion equation (Laplace equation) is solved on the lattice. The concentration field is initialized using the analytical linear profile between the bottom and top boundaries,

$$c(y) = c_{\text{bottom}} + \frac{y}{N-1}(c_{\text{top}} - c_{\text{bottom}}).$$

The following boundary conditions are applied:

- $c = c_{\text{bottom}}$ at the bottom boundary,
- $c = c_{\text{top}}$ at the top boundary,
- zero–flux conditions at the left and right boundaries,
- $c = 0$ at all lattice sites belonging to the cluster.

The Laplace equation is solved iteratively using the discrete update

$$c_{i,j}^{(k+1)} = \frac{1}{4}\left(c_{i-1,j}^{(k)} + c_{i+1,j}^{(k)} + c_{i,j-1}^{(k)} + c_{i,j+1}^{(k)}\right),$$

until the maximum change between successive iterations falls below a tolerance $\varepsilon = 10^{-5}$.

B. Growth candidates and probabilities

Growth candidates are defined as lattice sites that do not belong to the cluster but have at least one nearest neighbour (north, south, east, or west) that is part of the cluster. For each candidate (i,j) the growth probability is computed as

$$p_g(i,j) = \frac{c_{i,j}^{\eta}}{\sum_{\text{candidates}} c_{i,j}^{\eta}},$$

where η controls the morphology of the cluster.

C. Cluster growth algorithm

The cluster grows according to the following steps:

1) Solve the diffusion equation on the current domain.
2) Identify all candidate growth sites.
3) Compute the growth probabilities from the concentration field.
4) Randomly select one candidate according to p_g.
5) Add the selected site to the cluster.

Each iteration therefore increases the cluster by one node.

D. Optimized solver

To reduce the computational cost, an optimized solver based on the red–black Successive Over-Relaxation (SOR) scheme is also implemented. The lattice is divided into two checkerboard subsets that are updated alternately using

$$c_{i,j}^{(k+1)} = (1-\omega)c_{i,j}^{(k)} + \omega \bar{c}_{i,j},$$

where $\bar{c}_{i,j}$ is the average of the four neighbouring nodes and ω is the relaxation parameter. The value

$$\omega = \frac{2}{1+\sin(\pi/N)}$$

is used as an approximation of the optimal relaxation parameter. The solution from the previous growth step is used as the initial guess for the next diffusion solve.

E. Simulation parameters

Clusters are generated for different values of the morphology parameter

$$\eta \in \{0, 0.5, 1, 1.5, 2\},$$

typically on a 100×100 grid. In the optimized implementation larger grids up to 500×500 are also considered in order to evaluate the computational efficiency of the solver.

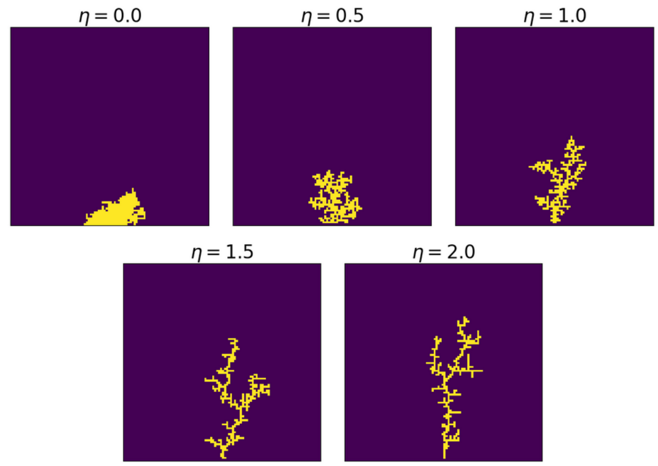

Fig. 1: Clusters generated with the diffusion-based DLA model for different values of the morphology parameter η on a 100×100 lattice. Yellow cells represent occupied lattice sites belonging to the cluster.

IV. Results and Discussion

Figure 1 illustrates the effect of the morphology parameter η on the resulting aggregates. For $\eta = 0$, all candidate sites have equal attachment probability and the cluster grows remaining in compact shape. As η increases, growth becomes increasingly biased toward sites with higher concentration, typically located at exposed tips. The aggregates therefore become progressively more branched. In particular, $\eta = 1$ corresponds to the classical DLA case, while larger values further enhance tip growth and produce highly ramified structures.

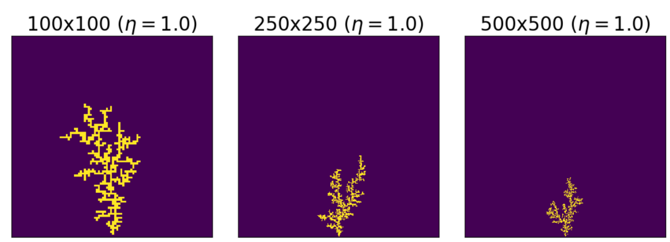

Fig. 2: Clusters obtained using the red–black SOR implementation for different grid sizes with $\eta = 1$.

To reduce the computational cost of repeatedly solving Laplace's equation during cluster growth, the SOR solver was reformulated using a red–black ordering. The lattice is divided into two checkerboard sublattices: red nodes are updated using the current black values, followed by an update of the black nodes using the newly computed red values. This gives a more efficient update while preserving convergence of SOR.

Figure 2 shows clusters obtained with the improved solver for increasing grid sizes. The overall morphology remains consistent across the different resolutions, indicating that the numerical implementation gives the same growth behaviour on larger domains.

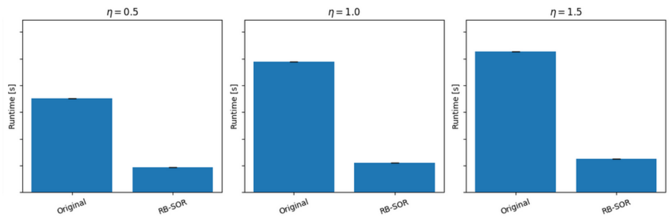

Fig. 3: Runtime comparison between the standard solver and the red–black SOR implementation for different values of η.

The performance gain of the improved solver is quantified in Fig. 3. For all tested values of η, the red–black SOR implementation consistently achieves a lower runtime than the original solver. This improvement arises from the structured update pattern, which leads to more efficient memory access and better suitability for vectorized computations.

An additional speedup comes from the fact that the cluster grows by only one lattice site at each step. As a consequence, the concentration field changes only slightly between successive solves of the diffusion equation. Using the previously converged solution as the initial guess therefore reduces the number of iterations required for convergence. Together with the red–black ordering, this significantly improves the computational efficiency.

EXERCISE 2

I. Introduction

In addition to the diffusion-based approach considered in the previous exercise, DLA can also be simulated using a Monte Carlo model based on random walkers. In this exercise we implement such a simulation on a two-dimensional lattice. Random walkers are released from the top boundary and move until they either become part of the cluster or leave the system. A sticking probability p_s controls attachment when a walker reaches a site neighboring the cluster. The aim is to investigate how the cluster morphology depends on the value of p_s.

II. Theory

In the classical formulation of diffusion-limited aggregation, cluster growth is governed by a diffusion field satisfying Laplace's equation. An alternative formulation models diffusion explicitly through random walkers: instead of solving the diffusion equation, particles perform random walks on the lattice, statistically reproducing diffusion.

When a walker reaches a site adjacent to the cluster it may attach and become part of the aggregate. In the classical Monte Carlo DLA model this occurs with probability one. More generally, a sticking probability p_s can be introduced: a walker attaches with probability p_s, otherwise the random walk continues. For $p_s = 1$ the classical DLA structure is obtained, while smaller values typically produce more compact aggregates.

III. Methods

A. Computational domain

The simulations are carried out on a square lattice of size 100×100. Each cell can be empty or belong to the cluster. The cluster is stored as a binary matrix $\mathbf{C}$ where

$$C_{i,j} = \begin{cases} 1 & \text{if the site belongs to the cluster} \\ 0 & \text{otherwise.} \end{cases}$$

The initial configuration consists of a single seed particle located at the bottom center of the domain.

B. Random walker dynamics

Particles are introduced one at a time. Each walker starts at a random horizontal position on the top boundary.

At every step the walker moves to one of the four neighboring lattice sites (up, down, left, or right), chosen with equal probability.

C. Boundary conditions

Different boundary conditions are applied in the horizontal and vertical directions.

Horizontally, periodic boundary conditions are used: when the walker crosses the left or right boundary it re-enters from the opposite side.

Vertically, the boundary conditions impose that, if the walker exits through the top or bottom boundary, the walker is removed and a new walker is released.

D. Cluster attachment rule

A walker may attach to the cluster when it occupies a site adjacent to it, meaning that one of its four nearest neighbors already belongs to the cluster.

When adjacency is detected, a random number r is drawn from the uniform distribution on $[0, 1)$. The walker attaches if $r < p_s$; otherwise, the random walk continues.

Walkers are not allowed to move into cells that already belong to the cluster.

E. Simulation procedure

The simulation proceeds by repeatedly releasing walkers until a desired number of particles has been attached.

The algorithm is:

1) Initialize the cluster with a single seed particle.
2) Release a walker at a random position on the top boundary.
3) Perform random walk steps.
4) Apply boundary conditions.
5) If the walker is adjacent to the cluster, attach it with probability p_s.
6) If the walker leaves the domain, discard it and release a new walker.
7) Repeat until the required number of particles has been added.

In the simulations the cluster grows until 700 particles have attached.

IV. Results

Figure 4 shows the cluster obtained with sticking probability $p_s = 1$, corresponding to the classical Monte Carlo formulation of DLA. The structure exhibits long branches, indicating a growth mainly at tips.

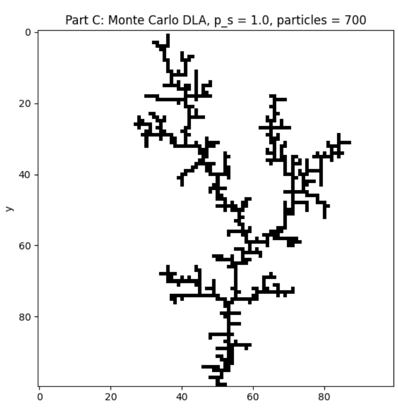

Fig. 4: Monte Carlo DLA cluster for $p_s = 1$ on a 100×100 lattice with 700 particles.

To investigate the effect of the sticking probability, additional simulations were performed for $p_s = 1.0$, 0.5, 0.2, and 0.05 (Fig. 5).

A clear transition occurs as p_s decreases. For $p_s = 1$ the cluster is sparse and strongly branched. For $p_s = 0.5$ the structure remains ramified but becomes slightly denser. For smaller values such as $p_s = 0.2$, branches thicken, and more compact clusters appear.

For very small sticking probability ($p_s = 0.05$), walkers often continue their motion after reaching the cluster and penetrate deeper towards the seed location making clusters significantly more compact.

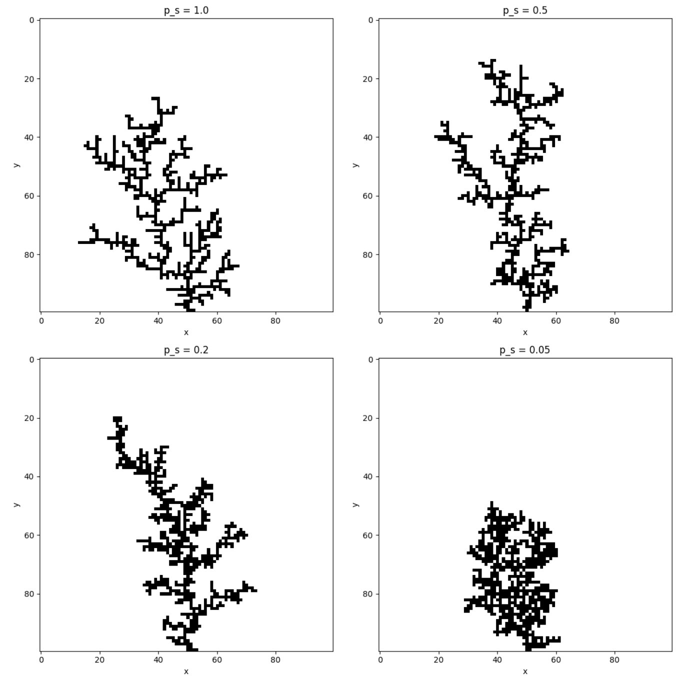

Fig. 5: Clusters generated for different sticking probabilities p_s. Each simulation was performed on a 100×100 grid with 700 particles.

V. Discussion and Conclusions

The simulations confirm that the sticking probability p_s strongly influences the morphology of the aggregates. Large values of p_s favor immediate attachment at the cluster boundary, leading to growth mainly at exposed tips and therefore to sparse, highly branched structures. When p_s decreases, walkers are more likely to continue their motion after first contact with the cluster and can penetrate deeper toward the interior, producing progressively more compact aggregates.

This behavior is consistent with the diffusion-based formulation studied in Exercise 2.1. Although the parameters are different, both models control the competition between tip growth and more uniform aggregation. In the diffusion model this effect is governed by the exponent η, while in the Monte Carlo formulation it is determined by the sticking probability p_s.

Overall, the Monte Carlo model provides a simple stochastic representation of diffusion-limited growth. Despite its simplicity, it reproduces the main morphological features observed in the diffusion-equation approach without requiring the repeated numerical solution of Laplace's equation.

EXERCISE 3

I. Introduction

This exercise studies the Gray–Scott model, which describes the interaction between two chemical species whose concentrations evolve in time and space. Despite the simplicity of the reactions, the model can generate a wide variety of spatial patterns from simple initial conditions.

The aim is to simulate the model in two spatial dimensions and investigate how the behavior changes when the parameters are varied and when noise is added to the initial conditions.

II. Theory

A. Gray–Scott reaction model

The Gray–Scott model describes the interaction between two chemical species U and V governed by the reactions

$$U + 2V \to 3V, \qquad V \to P.$$

Let $u(x, y, t)$ and $v(x, y, t)$ denote the concentrations of U and V. Their evolution is described by

$$\frac{\partial u}{\partial t} = D_u \nabla^2 u - uv^2 + f(1-u)$$

$$\frac{\partial v}{\partial t} = D_v \nabla^2 v + uv^2 - (f+k)v.$$

Here D_u and D_v are diffusion coefficients, f is the feed rate of U, and $f + k$ determines the decay rate of V.

B. Diffusion operator

The Laplacian models diffusion. In two spatial dimensions,

$$\nabla^2 u = \frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2}.$$

Diffusion smooths concentration gradients, while the nonlinear terms can amplify local variations.

III. Methods

A. Spatial discretization

The spatial domain is discretized on a uniform square grid of size $N \times N$ with spacing Δx, with indices $i, j = 0, \dots, N-1$.

The Laplacian is approximated using central finite differences

$$\nabla^2 u_{i,j} \approx \frac{u_{i+1,j} + u_{i-1,j} + u_{i,j+1} + u_{i,j-1} - 4u_{i,j}}{(\Delta x)^2},$$

and similarly for v.

B. Time discretization

According to the Method of Lines, time is discretized with the explicit Forward Euler scheme. The updates become

$$u_{i,j}^{n+1} = u_{i,j}^n + \Delta t \left(D_u \nabla^2 u_{i,j}^n - u_{i,j}^n (v_{i,j}^n)^2 + f(1 - u_{i,j}^n) \right),$$

$$v_{i,j}^{n+1} = v_{i,j}^n + \Delta t \left(D_v \nabla^2 v_{i,j}^n + u_{i,j}^n (v_{i,j}^n)^2 - (f+k) v_{i,j}^n \right).$$

C. Boundary conditions

Periodic boundary conditions are used in both spatial directions, so indices wrap around the grid, e.g. $u_{-1,j} = u_{N-1,j}, \quad u_{N,j} = u_{0,j}$.

D. Initial conditions

The field u is initialized uniformly as:

$$u_{i,j}(0) = 0.5.$$

The field v is initialized as zero except in a central square region:

$$v_{i,j}(0) = 0.25 \quad \text{for } 25 \le i \le 34,\ 25 \le j \le 34,$$

and $v_{i,j}(0) = 0$ elsewhere.

Optional small random perturbations are added to both fields to study the effect of noisy initial conditions.

E. Simulation parameters

We use the parameters suggested in the assignment:

- $\Delta t = 1, \quad \Delta x = 1;$
- $D_u = 0.16, \quad D_v = 0.08;$
- $f = 0.035, \quad k = 0.060.$

The simulations use $N = 60$ and run for 2000 time steps. Additional parameter pairs (f, k) are tested to study their influence on pattern formation.

F. Algorithm

The numerical procedure is:

1) Initialize the fields u and v on the $N \times N$ grid.
2) Optionally perturb the initial condition with uniform noise levels $[\texttt{None}, 0.01, 0.05, 0.1]$.
3) For each time step:
   a) compute the discrete Laplacians of u and v;
   b) evaluate the nonlinear term uv^2;
   c) update both fields using the Forward Euler scheme.
4) Repeat the simulations for different (f, k) parameter sets and noise levels, storing snapshots every 10 time steps for visualization.

IV. Results and Discussion

Simulations were performed for different noise levels and parameter combinations. Results are shown for the concentration field v, since the patterns of u are complementary and display similar structures.

Figure 6 shows the initial and final concentration fields of v for the reference parameter set $(f, k) = (0.035, 0.060)$ without noise. Starting from a localized initial condition, the system evolves toward a nontrivial spatial pattern.

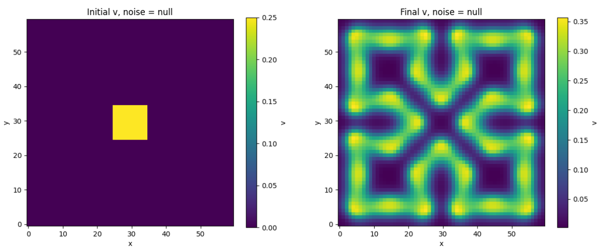

Fig. 6: Initial and final concentration fields of v for the reference parameter set $(f, k) = (0.035, 0.060)$ without noise.

To evaluate the effect of perturbations in the initial condition, additional simulations were performed with different noise amplitudes. Figure 7 compares the final v field for the reference parameters with no noise and with maximum noise (i.e. noise ranging between ± 0.1). Without noise the pattern remains highly symmetric due to the symmetric initial condition. Introducing noise breaks this symmetry, although the overall structure of the pattern remains similar. Smaller noise levels (not shown) produce only minor deviations.

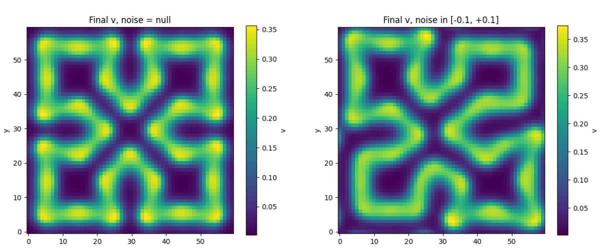

Fig. 7: Comparison of the final concentration field of v for $(f, k) = (0.035, 0.060)$ with no noise and with maximum noise.

The influence of the parameters was further investigated by varying f and k in the absence of noise. Figure 8 shows the resulting patterns for the tested parameter sets. Even small parameter changes lead to significantly different structures, highlighting the strong sensitivity of the Gray–Scott model to parameter variations while preserving the overall symmetry of the domain.

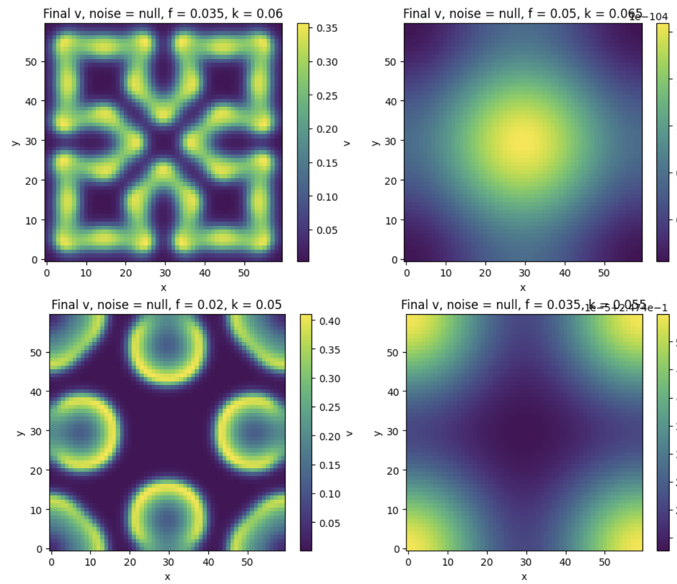

Fig. 8: Final concentration field of v for the parameter combinations $(0.035, 0.060)$, $(0.050, 0.065)$, $(0.020, 0.050)$ and $(0.035, 0.055)$ without noise.

Animations show how spatial patterns gradually emerge and evolve. Within the simulated time horizon (2000 steps), most parameter combinations are still evolving, while the baseline case $(f, k) = (0.035, 0.060)$ approaches a more stable configuration.